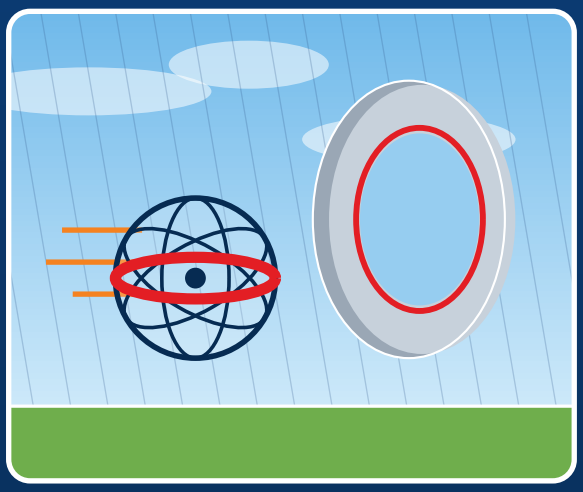
GALGOTIAS UNIVERSITY | SOEngg Department of Electrical, Electronics & Communication Engineering | GALGOTIAS UNIVERSITY G.C.D.I.S.

National Drone Soccer League

Rule Book | Class D20



Highlights

- Teams of 2–5 members
- Drone ball Ø 20 cm
- Max drone weight 150 g
- Custom-made cages allowed
- 3 sets × 3 min, penalty kicks



Standard cage 8 × 4 × 3 m

Goal: Ø 30 cm ring | Pilot areas 2 m each end

ARENA LAYOUT (TOP VIEW)

Match Workflow

Register → Inspect Drones → Toss & Ready → Play 3 Sets → Penalty Kicks → Winner

REGISTRATION

Registration fee
Rs. 1500/-

Security deposit
None

Own drone
Allowed

LEAGUE DETAILS

Team size
2–5 members

Max drone weight
150 g

Drone ball
Ø 20 cm

Game
3 sets × 3 min

ARENA

Cage
Custom-made allowed

Standard size
8 × 4 × 3 m

Goal
Ø 30 cm ring

Pilot areas
2 m at each end

Learn Build Fly

GALGOTIAS CENTRE FOR DRONE INTELLIGENCE & SIMULATION (GCDIS)

Department of Electrical, Electronics & Communication Engineering
School of Engineering, Galgotias University

2–5	Ø 20 cm	150 g	3 × 3 min	5 min	Rs. 1500
Members per team	Drone ball	Max drone weight	Sets per game	Repair between sets	Registration fee

1 TEAMS & REGISTRATION

- Each team registers **2–5 members**. Every member pilots one drone ball, so a team flies one ball per pilot (maximum 5).
- Registration fee: **Rs. 1500/-**. There is no security deposit.
- At least **2 pilots** must be seated before a game, otherwise the game is forfeited. Fewer than 2 before set 2 or 3 means that set is lost.
- Substitutions only between sets, from the registered roster. Tell the referee before the ball enters the cage; the substitute's ball is inspected.
- Only registered pilots may stay in the pilot area.

2 ARENA (CAGE)

- Custom-made cages are allowed, provided their dimensions match the specification: **8 m long × 4 m wide × 3 m high**.
- Floor flat, free of obstacles and preferably not hard. Mark a half line across the middle of the long side, and takeoff points (a line or five dots) **1.5 m** from each short side.
- A **2 m** wide pilot area sits at each short end, separated from the repair area by a boundary marker.
- Walls of net or wire so the game stays visible. A net must not deflect more than **10 cm** when hit at 3 m/s; wires run vertically, **1–2 mm** thick, **10 cm** apart. A ball must never leave the cage.

3 GOAL

- Ring: inner Ø **30 ± 1 cm**, outer Ø **50 ± 1 cm**, **10 cm** thick. Both goals must be identical.
- Placed **1.5 m** in from the centre of each short side towards the half line, with the ring centre **1.8–2.2 m** above the floor. Hung from one or two attachment points or fixed on posts, facing the centre without swaying, and secured so it cannot fall or change shape.
- White and red LEDs, switched manually from outside the cage. The ring turns **red** for an approved goal and back to white afterwards.

4 DRONE BALL & EQUIPMENT

- Round exoskeleton, Ø **20 ± 1 cm**. **Maximum weight 150 g** in play for every drone ball. No single open surface larger than **70 cm²**; the shell must not break in a way that endangers people.
- Every ball is weighed and inspected before the first match, and the referee may re-check at any time.
- Team LED ring in **red** or **blue**, visible from all sides. The Striker also carries a **green** tag and LED. No other colours that confuse team or position. If the Striker's tag falls off and opponents cannot identify it, goals scored afterwards do not count.
- One controller per pilot, bound to that pilot's ball. Spare batteries and spare balls are allowed (spare-ball batteries removed and stored apart). FPV is optional.
- Radio frequencies must follow local law and must never disturb another team's control.
- **Banned:** lights that dazzle opponents, jammers or transmitters, distracting audio devices, and modifications made only to gain an unfair advantage.

5 MATCH FORMAT

- A game is **3 sets of 3 minutes**. No time-outs. The coin-toss winner picks the pilot area (left or right), fixed for all sets unless the referee rules it unfair.
- Balls wait at the takeoff points and pilots stand in the pilot area. Whistle: **1 long = ready** (at least 10 s before start), **1 short = start** (given unpredictably), **short + long = end of set**. Moving a ball before the start signal earns a warning.
- **Repair time:** at least 5 minutes between sets, counted from when all pilots and balls have left the cage. It is not extended just because a team is not ready; LED and tag fixes may be made at the referee's request.
- The referee may pause play for safety or arena faults; score and time are restored on resume.

A set is void if the cage is badly damaged or play cannot resume within one hour.

- Abandoning a game or deliberately delaying it is a game loss.

6 SCORING, OFFSIDE & DEFENCE

- **Goal:** the Striker's ball passes completely through the opponent's goal from front to back. Only **Strikers** (maximum two per team) can score; every other pilot defends.
- **No goal** if the ball rebounds without clearing the ring, passes back to front, or the team is offside.
- **Offside:** after a Striker passes through the opponent's goal in either direction, the team cannot score again until every one of its balls in the opponent's half has returned behind the half line. If a ball there is uncontrollable, offside stays until its pilot gives up the set and puts the controller down.
- **Defence:** defenders may fly through their own goal from front to back only. Entering from the back, even slightly, or entering from the front with more than half the ball and coming back out, is a **reverse entry**. A ball that pokes past the back of the goal must retreat to its defending position.

7 PENALTY KICKS

- A penalty kick is awarded for: consecutive goals scored while offside, a reverse entry into a team's own goal, or **two accumulated warnings** (they carry across sets and reset every game).
- After each set, penalties of both teams are offset and the difference is awarded to the other team. All penalties are still recorded.
- A kick is one-on-one, **Striker vs Keeper**, for **5 seconds** from the referee's signal. The Striker starts at the half line, the Keeper (one defender) below its goal. Several goals may be scored in the 5 seconds; offside rules still apply.
- **Group stage:** every awarded kick must be taken (it counts for goal difference). **Knockout stage:** the awarded team may waive it. The set ends after the kick, followed by 5 minutes of repair time.

8 WINNING

- More goals in a set wins the set; equal goals (including 0–0) is a drawn set.
- After three sets the team with more set wins is the winner: two wins, or one win and two draws.
- Still level: one **overtime set** in the same format, then a **penalty shoot-out** of 3 kicks per team (pilots may repeat in small teams) in a fixed order, then sudden death: the first team to score while the other misses wins. If a competition allows draws, there is no overtime or shoot-out.

9 VIOLATIONS

- **Warning:** a non-pilot staying in the pilot area; minor unsportsmanlike behaviour; moving arena equipment without permission to gain an advantage; moving a ball before the start signal; ignoring a legitimate referee instruction.
- **Set loss:** someone outside the set controlling a ball in play; deliberately changing ball colours; moving a ball by hand, foot or equipment instead of radio control; repeating the same warning to delay or protest; banned equipment that affected play.
- **Game loss:** using a drone to threaten or endanger anyone; serious unsportsmanlike conduct towards officials, opponents or spectators; an unregistered person flying or being included.

10 OFFICIALS & DECISIONS

- Every game has one **head referee** and two assistant referees (up to four: score referees and penalty referees). Assistants judge goals, offside and penalties; if they disagree, the head referee decides. Head referee decisions are final.
- The head referee may stop the game, remove a pilot for safety, and review the official recording to correct a decision before play resumes. Private footage is never accepted.
- **Assistant signals:** flag raised and goal ring turns red = goal approved; flags raised in a V and ring stays white = goal disapproved; flag held out sideways = offside return declared; flag lowered and ring back to white = return completed.